

imperfect depictions of economic reality. In 2008, economists' models failed to see the financial crisis that was already unfolding.¹⁵¹ None of their models anticipated the rise of nonbank lending and mortgages that eschewed traditional standards like credit checks, employment verification, and loan-to-value ratios. They did not even consider the possibility that *housing prices might go down!* These models were in part why there was such significant delay in recognizing and responding to the GFC—the worst recession since the Great Depression.

Our faith in financial models should already have been tempered by events a decade earlier, when the hedge fund Long-term Capital Management collapsed. Their complex models¹⁵²—created by two brilliant Nobel laureates—never anticipated that Russia would default on its debt.

When the world changes, but our models don't, we run into trouble:

- Models for securitized mortgages and related derivatives (CDOs, CMOs, CLOs, etc.) never anticipated housing prices going down. This was a major factor leading up to the GFC.
- Value investing models, championed by Benjamin Graham and Warren Buffett, ran into trouble in the 2010s because they failed to anticipate low inflation, quantitative easing (QE), and zero interest rate policy (ZIRP) by the Federal Reserve.
- In 2020, models went mad due to the pandemic lockdowns. With demand for crude oil crashing, and storage space for oil completely filled up, oil futures briefly traded at a negative \$40 a barrel—meaning those holding oil had to pay to have it taken off their hands.
- Before 2020, the Bureau of Labor Statistics never had to cope with 20% of the workforce filing for unemployment claims in a single month. You can be sure that those models did not anticipate that.
- Negative interest rates around the world in the 2010s blew up a lot of models of inflation and yield, including those used by the Federal